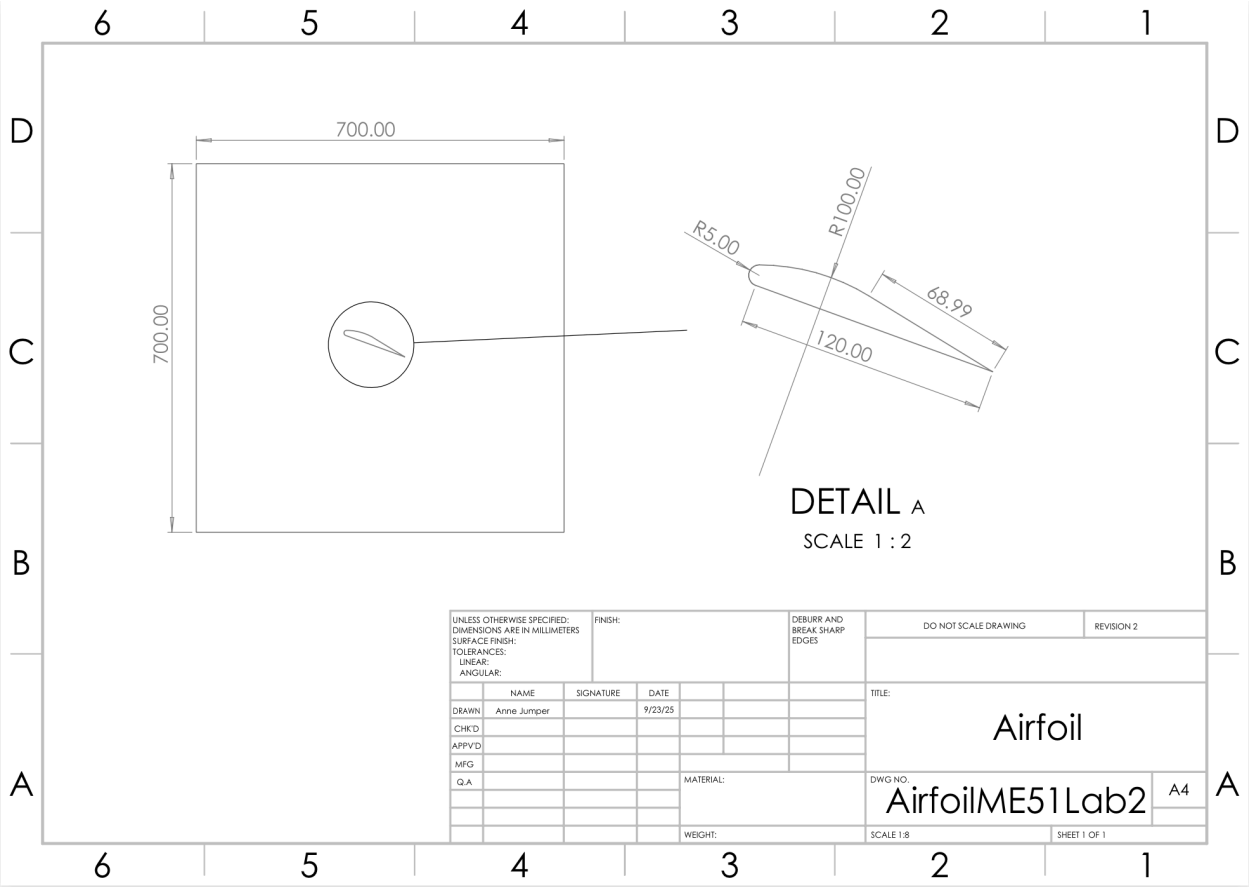
Lab 2: 2D CFD Simulation in COMSOL

Anne Jumper

Deliverable 1: Technical Drawing of Airfoil Geometry



This airfoil was drawn first using a 120 mm segment tilted 20 degrees clockwise. A circle of radius 5 mm is placed at one end and connected to it is a 3 point arc of radius 100 mm. The shape is closed using a line and the trim entities tool is used to remove the part of the circle within the airfoil boundary. Finally, tangent relations are applied at the points connecting the two arcs and the 100 mm arc and line segment.

Deliverable 2: Inlet Velocity and Reynold's Number Calculation

Inlet Velocity: $U_0 = 0.08$ m/s

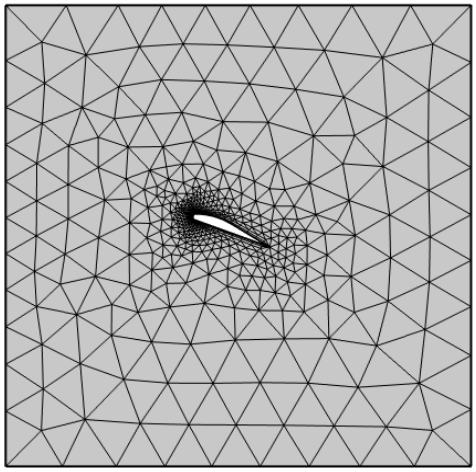
Reynold's Number Calculation:
Velocity (V) = 0.08 m/s
Length of Airfoil (c) ~ 0.12 m + 0.005 m = 0.125 m
Kinematic Viscosity of Air at Room Temperature (ν) = 1.52×10^{-5} m^2/s

$$R = \frac{Vc}{\nu} = \frac{(0.08\ m/s)(0.125\ m)}{(1.52 \times 10^{-5}\ m^2/s)} = 657.8947368$$

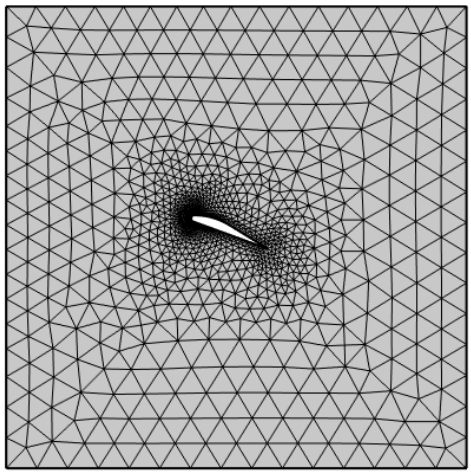
Deliverable 3: Comparison of Mesh Sizes

Entire Domain:

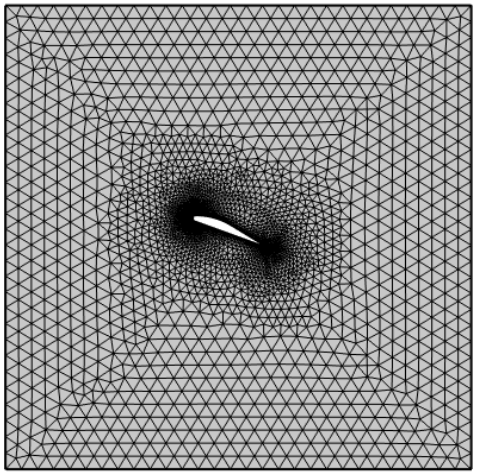
Coarser



Normal

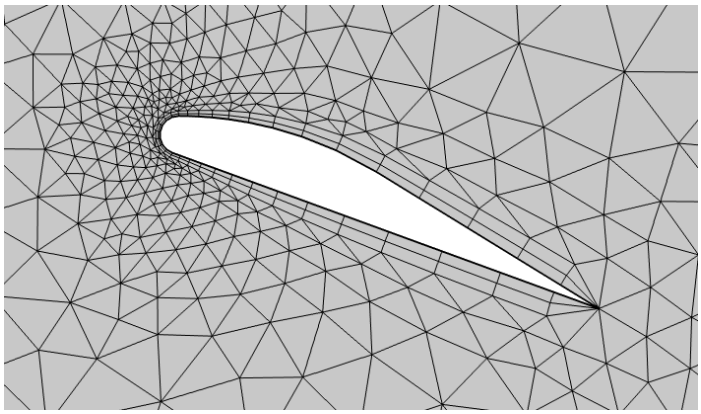


Finer

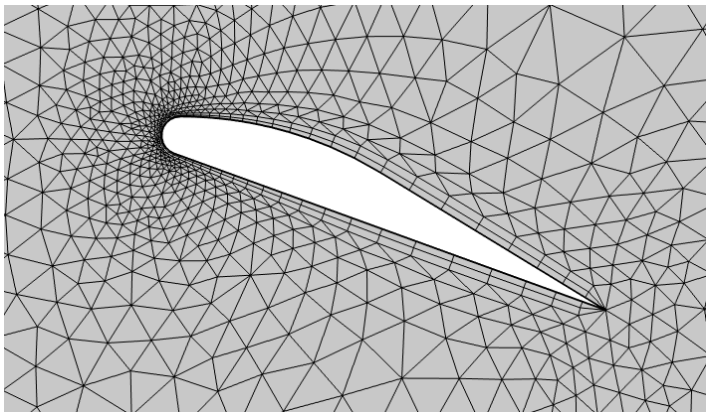


Around Airfoil:

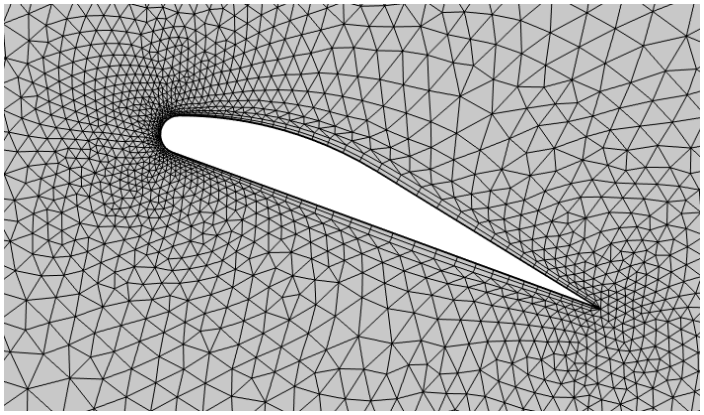
Coarser



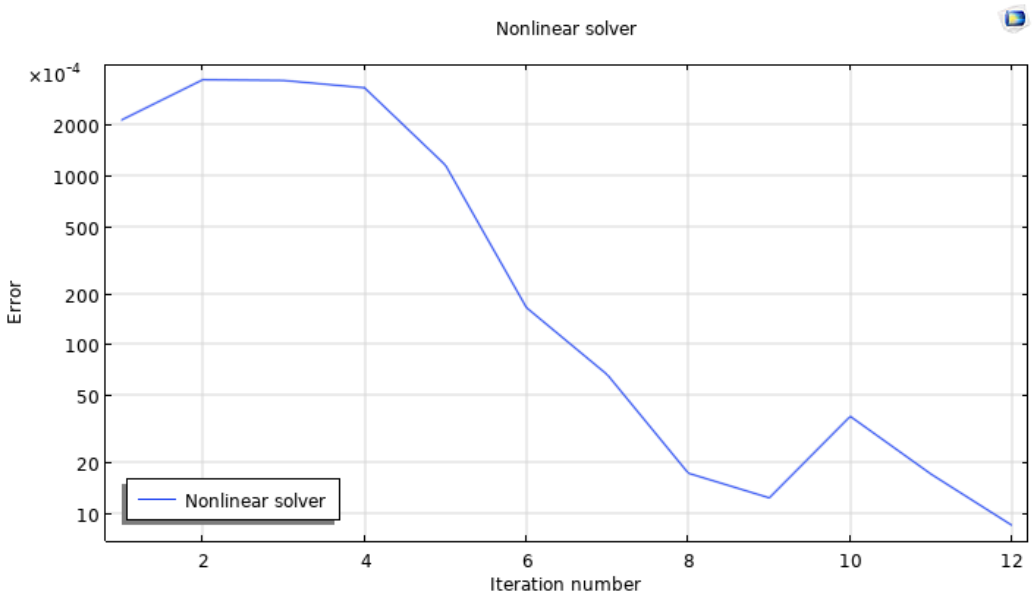
Normal



Finer

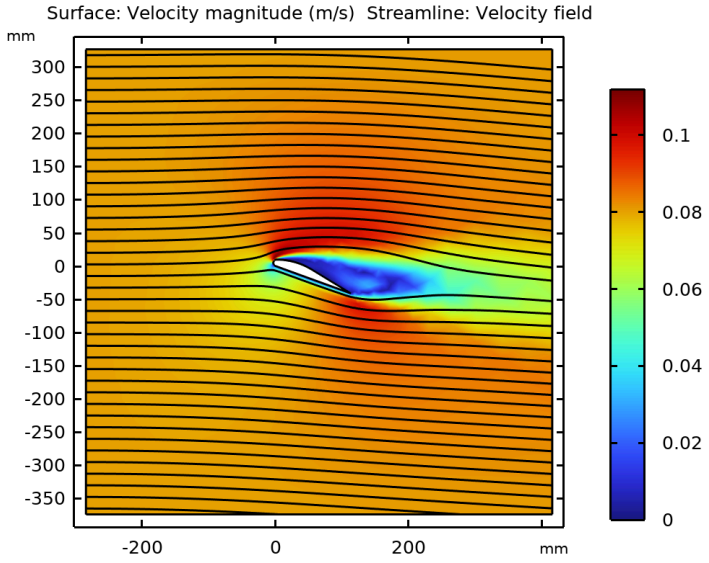


Deliverable 4: Computation Time and Solution Convergence
Solution Time: 2s
Convergence Plot:

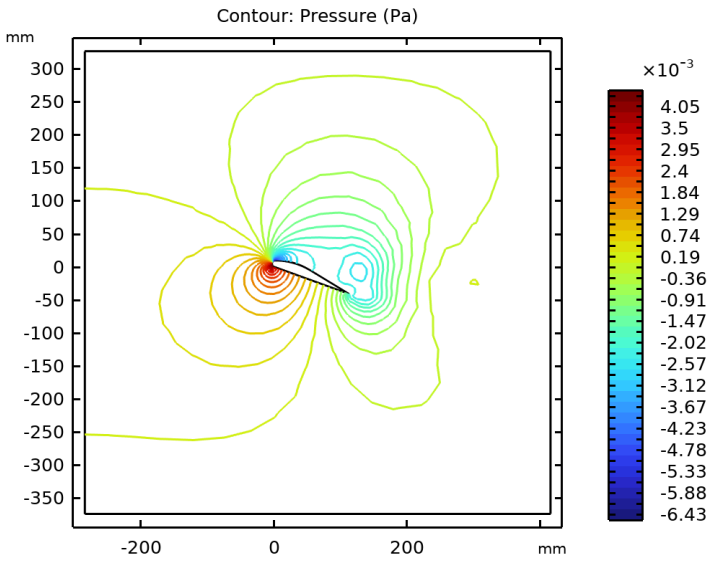


Deliverable 5: Data Analysis

Velocity Magnitude (m/s) with Streamlines



Pressure (Pa) Field



The flow is fast around the top and bottom edges of the airfoil, indicated by the dark red regions. Flow is faster; however, at the top edge compared to the bottom. Probing reveals the maximum velocity to be 0.111960 m/s at the top edge. Flow is slowest behind the airfoil on the right side, where the airfoil itself shields the region from flow. Probing reveals the minimum velocity to be 0 m/s along the edge of the airfoil due to the no-slip condition. Looking at the streamlines, flow is mostly horizontal, particularly in regions far from the airfoil. There are significant upward vertical components to velocity near the leading edge where flow is diverted to go around the airfoil (seen as bending streamlines) and downward vertical components after the tail of the airfoil where the air no longer has an object to divert flow. Something of note is that the regions of high and low pressure closely correspond to regions of high and low velocity magnitude. Higher positive pressures are seen near the nose of the airstrip (high velocity region) and lower negative ones are seen behind in the wake of the airstrip (low velocity region).

Deliverable 6: Drag and Lift Forces

Quantity	**Value (N/m)**
Viscous Force, x-direction	3.7814E-5
Viscous Force, y-direction	1.7914E-6
Pressure Force, x-direction	1.9624E-4
Pressure Force, y-direction	4.1876E-4

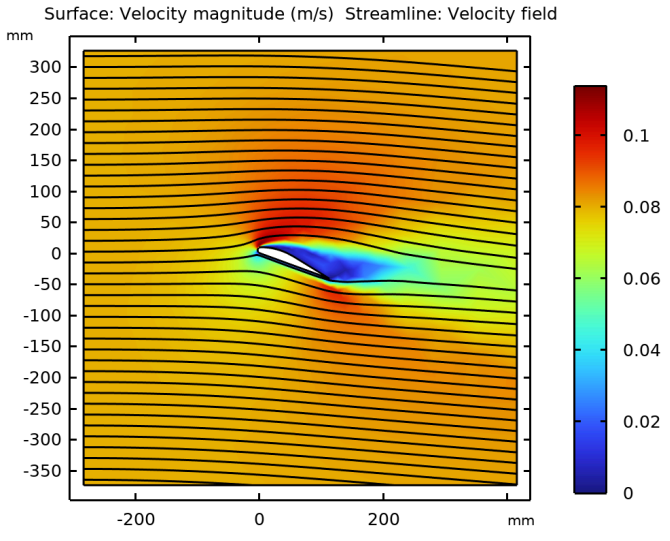
Looking at the magnitudes of the forces, mine have a similar order of magnitude to those given: the pressure forces significantly higher than the viscous forces, the pressure in the y-direction is smaller than that in the x direction, and the viscous force has less magnitude in the y-direction than x-direction. However, all of my values are slightly smaller in magnitude. Much of this could be explained by different inlet velocities and the smaller incline angle of my airfoil which translates to less force being exerted on my airfoil 'trying' to make it horizontal (in line with the flow so that less air is diverted, less drag).

The most notable difference is that the viscous force in the y-direction is positive while that from the example given is negative. This is likely due to the fact that my airfoil is only tilted about 20 degrees while the one given is tilted 30 degrees. The most 'efficient' flow with fewer high pressure/velocity regions (less drag) occurs when the airfoil isn't inclined at all, so the more inclined airfoil experiences higher force to return it to this 'ideal' state. Additionally, the camber of my airfoil increases air flow speeds on the upper edge, translating to less downward force on my airfoil compared to the uncambered airfoil due reduced pressure above relative to below.

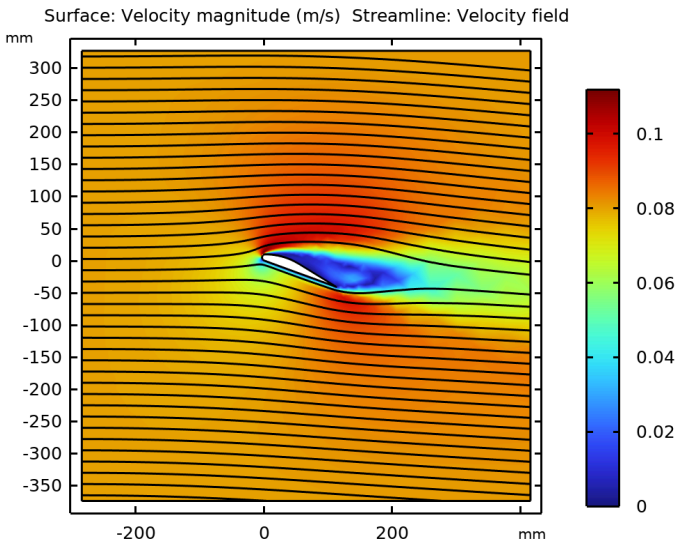
Overall, I think the answer I got was reasonable considering my altered incline angle and asymmetrical design.

Deliverable 7: Examination of Grid Convergence

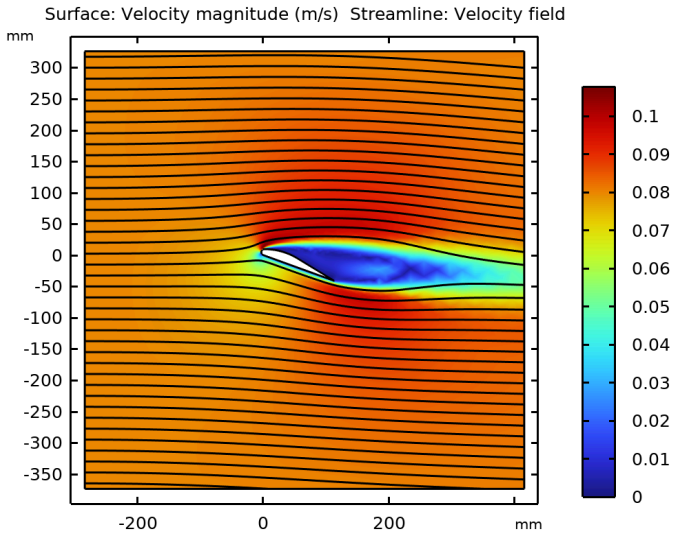
Coarser



Normal



Finer



As the mesh becomes finer and finer, the resolution of the plots increases. The coarser mesh produces a plot where color regions are less defined and less precise. On the other hand, the finer mesh has more concentrated and precise regions of different colors, showing regions of low and high velocity more distinctly. The color contours are sharper and the boundary between low and high velocity is more well-defined.

Mesh Size	**Number of Mesh Elements**	**Computation Time (seconds)**
Coarser	930	0
Normal	2302	1
Finer	5761	1

Deciding which mesh to use is a trade-off between accuracy and computing power. When results need to be extremely precise and computation efficiency is of less concern, it is best to use a finer mesh. For more efficient results where the precision of numbers is less important than a general idea of the surface plot, a coarser mesh may be desirable. By computing results for multiple mesh sizes, you can compare the convergence plots to see if it has leveled out sufficiently, indicating mesh independence and result accuracy. COMSOL also lets you set the convergence tolerance which is another way to check if results are ‘good enough’.

Deliverable 8: Reflection

My process began with doing some research on what a typical airfoil looks like so I could better understand how one of these behaves in a real world environment using my design. I ended up designing a cambered airfoil which is more curved at the top than the bottom. This ended up causing the most confusion for me since it forced me to think about what would cause the discrepancies in my lift and drag forces from those given. I learned that fluid moves more quickly along the curved upper edge (which when observing my velocity surface plots also became clear), creating lower pressures (more air displacement) above the airfoil compared to below. This generates additional lift, explaining my non-negative lift forces which were causing me confusion. I ended up re-running my simulation several times with more refined mesh sizes, thinking that poor solution convergence was causing inaccurate calculations of the drag and lift force, but after researching airfoils a little more, I was able to eliminate much of my confusion. In the end, reconciling my results was probably the hardest part of this lab for me. I still have a lot of questions about the camber of airfoils and how these are designed to optimize lift without causing significant drag in plane wings, propeller/turbine blades, and more. I hope to learn more about this in the future. My experience using COMSOL was a very positive one this time. I have used COMSOL in the past and have had every error imaginable, so I feel like my troubleshooting skills with the software have really improved which has made the labs run a lot more smoothly. Another issue I had was with getting an accurate computation time since once I computed a solution for one mesh, COMSOL would recompute the same solution from memory, taking less computation time. I had to learn how to clear the solution and mesh memory which was new for me.